

Arnav Bhilwariya

+91-7417296760 | arnavbhilwariya0408@gmail.com | linkedin.com/in/arnavli04 | github.com/arnav-144p

SUMMARY

ML engineering student with open source contributions to PyTorch, HuggingFace, and Stanford DSPy. Experienced in PyTorch, scikit-learn, LightGBM, FastAPI, and end-to-end ML pipelines. Focused on model deployment, feature engineering, and production-ready backend APIs.

EDUCATION

National Institute of Technology, Jalandhar

2023 – 2027

B.Tech in Information Technology | CGPA: 7.2 / 10

EXPERIENCE

AD Infocom Systems

July 2025 – August 2025

Software Engineer Intern – ML Systems

- Automated **data preprocessing and feature engineering pipelines** using Python scripting, reducing manual overhead by ~35% and cutting ML model training setup time by ~40%
- Shipped **3+ FastAPI endpoints** for real-time ML model inference; integrated predictions into client-facing REST APIs with Git-based version control, reducing average inference latency

OPEN SOURCE CONTRIBUTIONS

pytorch/torchTitan | [PR #3493](#) | *Merged*

2026

- Resolved broken CI flavor for **FSDP+TP+PP+compile** combination; unblocked distributed training test suite for core PyTorch training framework

BerriAI/litellm | [PR #29753](#) | *Merged*

2026

- Patched **proxy SSO role mapping misconfiguration** (`generic_role_mappings.default_role` → `generic_role_mappings.default_role`) in `ui.sso.py`; eliminated silent auth pipeline failure

stanfordnlp/dspy | [PR #9908](#) | *Merged*

2026

- Refactored **inline code comments** across core DSPy modules; improved contributor onboarding clarity for Stanford's LLM programming framework

PROJECTS

Goal-Directed Active Vision System | *PyTorch, CLIP, IRL* | [GitHub](#)

2026

- Trained **goal-directed visual search agent** using CLIP ViT-L/14 and Inverse Reinforcement Learning on COCO-Search18 (10K+ images); achieved **97.8% vs 88.5% passive baseline** on target localization accuracy
- Engineered **active vision pipeline** with PyTorch and OpenCV: frame sampling → CLIP feature extraction → IRL-based policy learning → fixation prediction evaluation

Retail Demand Forecasting | *Machine Learning, Time Series* | [GitHub](#)

2026

- Forecasted sales across 50+ stores and 3,000+ items using LightGBM; achieved **MAE 1.38 and RMSE 2.01**, outperforming ARIMA baseline by **22%**
- Implemented end-to-end **ML data pipeline**: ingestion → feature engineering (lag, rolling, date features) → model training → evaluation using scikit-learn and Pandas

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL

ML / AI: PyTorch, HuggingFace Transformers, Scikit-learn, TensorFlow, Keras, LightGBM, CLIP, NumPy, Pandas, Matplotlib

Concepts: Machine Learning, NLP, Feature Engineering, Time Series Forecasting, Model Deployment, Data Pipeline Design, Model Evaluation

Backend & APIs: FastAPI, Flask, Django, Node.js, REST API Design

Tools & Cloud: Git, GitHub, MLflow, AWS (EC2), Google Colab, Linux